

Generation Interconnection System Impact Study Report

For Queue Project Q19-1271

Tenaya 138 kV

April 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1271. The Interconnection Customer proposes a 203.7 MW solar photovoltaic facility interconnecting at the Tenaya 138 kV substation in the West Virginia area. The request was received on August 21, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 391 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,183,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MUD FORK to CHAUNCEY TAP 138 KV
Contingency: BARTLETTS FERRY to MACOMB 138 KV
Loading: 109.0% (Rating: 180 MVA)
DFAX: 6.2%
Pre-project loading: 101.9%
2. Facility: HEWETT to SAPPHO 138 KV L
Contingency: TRISTATE to VALEMOUNT 345 KV
Loading: 106.7% (Rating: 241 MVA)
DFAX: 15.1%
Pre-project loading: 94.0%
3. Facility: LOGAN to CHAUNCEY TAP 138 KV
Contingency: BARTLETTS FERRY to MACOMB 138 KV
Loading: 105.6% (Rating: 181 MVA)
DFAX: 6.2%
Pre-project loading: 98.6%
4. Facility: WATSON to MARTINEZ 138 KV
Contingency: HUBBARDSTOWN SUBSTATION to TRISTATE SUBSTATION 345 KV line
Loading: 105.5% (Rating: 262 MVA)
DFAX: 7.9%
Pre-project loading: 99.3%
5. Facility: WATSON to BUTTE PIPELINE 138 KV L
Contingency: HUBBARDSTOWN to TRISTATE 345 KV
Loading: 104.8% (Rating: 259 MVA)
DFAX: 7.9%
Pre-project loading: 98.7%
6. Facility: BUTTE PIPELINE to FIG GARDEN 138 KV
Contingency: HUBBARDSTOWN to TRISTATE 345 KV
Loading: 104.0% (Rating: 255 MVA)

- DFAX: 7.9%
Pre-project loading: 97.7%
7. Facility: MUD FORK to FORT MEADE 138 KV
Contingency: MACOMB to BARTLETTS FERRY 138 KV
Loading: 103.6% (Rating: 196 MVA)
DFAX: 6.2%
Pre-project loading: 97.1%
8. Facility: NORTHWEST BELOIT 230-1 kV XFMR
Contingency: TRISTATE to VALEMOUNT 345 KV
Loading: 103.1% (Rating: 513 MVA)
DFAX: 6.2%
Pre-project loading: 100.7%
9. Facility: NORTHWEST BELOIT 345/1 kV XFMR
Contingency: VAEMOUNT SUBSTATION to TRISTATE 345 KV
Loading: 103.1% (Rating: 513 MVA)
DFAX: 6.2%
Pre-project loading: 100.7%
10. Facility: HEWETT to MACOMB 138 KV
Contingency: VALEMOUNT to TRISTATE 345 KV Line
Loading: 102.3% (Rating: 342 MVA)
DFAX: 23.1%
Pre-project loading: 88.5%
11. Facility: BARTLETTS FRRY to MACOMB 138 KV line
Contingency: VALEMOUNT to TRISTATE 345 KV
Loading: 102.2% (Rating: 344 MVA)
DFAX: 23.1%
Pre-project loading: 88.5%
12. Facility: FORT MEADE to EL PATIO 138 KV line
Contingency: MACOMB to BARTLETTS FERRY SUBSTATION 138 KV
Loading: 102.0% (Rating: 202 MVA)
DFAX: 6.2%
Pre-project loading: 95.7%
13. Facility: EAST ZACHA to DWORSHAK 138 KV L
Contingency: BARTLETTS FERRY SUBSTATION to MACOMB 138 KV
Loading: 100.9% (Rating: 211 MVA)
DFAX: 6.2%
Pre-project loading: 94.9%
14. Facility: DWORSHAK to EL PATIO 138 KV Line
Contingency: MACOMB to BARTLETTS FERRY 138 KV L
Loading: 100.7% (Rating: 209 MVA)
DFAX: 6.2%
Pre-project loading: 94.7%
15. Facility: EAST ZACHA to KAW 138 KV
Contingency: BARTLETTS FERRY to MACOMB 138 KV line
Loading: 100.1% (Rating: 217 MVA)
DFAX: 6.2%
Pre-project loading: 94.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MUD FORK to CHAUNCEY TAP 138 KV	Reconductor line	\$1,897,000	100% (\$1,897,000)
HEWETT to SAPPHO 138 KV L	Reconductor line	\$2,406,000	89% (\$2,136,000)
LOGAN to CHAUNCEY TAP 138 KV	Reconductor line	\$2,443,000	100% (\$2,443,000)
WATSON to MARTINEZ 138 KV	Reconductor line	\$598,000	14% (\$81,000)
WATSON to BUTTE PIPELINE 138 KV	Reconductor line	\$1,167,000	14% (\$158,000)
BUTTE PIPELINE to FIG GARDEN 138 KV	Reconductor line	\$1,268,000	14% (\$171,000)
FT MEADE to MUD FORK 138 KV	Reconductor line	\$1,016,000	100% (\$1,016,000)
NORTHWEST BELOIT 230-1 kV XFMR	Replace transformer with higher-rated unit	\$6,134,000	5% (\$293,000)
NORTHWEST BELOIT 345/1 kV XFMR	Replace transformer with higher-rated unit	\$21,808,000	5% (\$1,041,000)
HEWETT SUBSTATION to MACOMB 138 KV	Reconductor line	\$1,952,000	85% (\$1,664,000)
BARTLETTS FERRY to MACOMB SUBSTATION 138 KV	Reconductor line	\$6,232,000	85% (\$5,313,000)
EL PATO to FT MEADE SUBSTATION 138 KV	Reconductor line	\$505,000	100% (\$505,000)
EAST ZACHA to DWORSHAK 138 KV L	Reconductor line	\$1,280,000	100% (\$1,280,000)
DWORSHAK to EL PATIO 138 KV	Reconductor line	\$1,001,000	100% (\$1,001,000)
E ZACHA to KAW 138 KV line	Reconductor line	\$4,184,000	100% (\$4,184,000)
	Total allocated to project		\$23,183,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.